📘 STORY: “Aarav’s Journey into AI & RAG”

👨‍🎓 Chapter 1: The Confused Beginner

Aarav was a second-year computer science student from Nepal. He knew basic programming in JavaScript and had built small projects like a to-do app and a weather app. But deep inside, he felt stuck.

Everywhere on the internet, he kept seeing words like:

- AI
- Machine Learning
- LLM
- RAG

He tried watching YouTube tutorials, but most of them were too fast or too theoretical. He would start a video, get confused, and quit halfway.

One day, he asked himself:

“Why does AI sometimes give wrong answers even when it sounds confident?”

That question changed everything.

📘 Chapter 2: Discovery of the Problem

Aarav learned that normal AI models (LLMs) don’t “know” everything. They only generate answers based on what they were trained on.

He discovered that:

- AI has **knowledge cutoff**
- AI can **hallucinate (make things up)**

This made him realize that:

“AI is like a smart student who sometimes guesses answers.”

This idea is actually true — traditional models rely only on stored knowledge, which can be outdated or incomplete

📘 Chapter 3: First Encounter with RAG

One day, Aarav found a blog explaining something called **Retrieval-Augmented Generation (RAG)**.

It explained:

Instead of answering from memory, AI first searches documents and then answers.

He understood it like this:

- Normal AI = Closed book exam
- RAG AI = Open book exam

This analogy helped him a lot, because RAG allows models to fetch real data before answering

📘 Chapter 4: First RAG Project (Failure 😅)

Aarav decided to build his first RAG project.

His idea:

👉 Upload PDFs → Ask questions → Get answers

What he did:

- Loaded PDF files

- Used embeddings
- Stored in vector DB
- Connected to LLM

Problem:

❌ Answers were wrong
❌ Retrieval was bad
❌ Sometimes irrelevant context

He got frustrated and thought:
"Maybe I'm not smart enough for this..."

📘 Chapter 5: Breakthrough Moment

After struggling for days, Aarav realized his mistakes:

❌ Mistakes:

- Large chunks (too big)
- No proper cleaning
- Bad query handling

✅ Fixes:

- Chunk size = 500–1000 tokens
- Added overlap
- Better prompt template

Then suddenly...
👉 His RAG started working.
Answers became:

- Relevant
- Context-based
- Accurate

This is exactly how RAG works — it retrieves relevant data first and then generates answers from that context

📘 Chapter 6: Real Project Idea (Jobmate AI 🔥)

Aarav got a startup idea:
👉 "What if I build AI for job seekers?"
He created:

💡 Features:

- Resume upload
- AI scoring
- Job matching
- Q&A from job descriptions

How RAG helped:

- Reads resumes
- Matches with job data
- Gives personalized suggestions

RAG allows systems to use external data instead of only training knowledge, making results more accurate and dynamic

📘 Chapter 7: From Learner to Builder

After 3 months:

- Aarav built a working AI system
- He understood embeddings, retrieval, prompts
- He stopped blindly following tutorials

He realized:
"AI is not magic. It's just systems working together."

📘 Chapter 8: Final Lesson

Aarav now teaches juniors:
"Don't try to learn everything. Build small, fail fast, fix, repeat."
And his favorite quote:
"RAG is not about AI being smart…
it's about giving AI the right information."